

Consumer Acceptance and Continued Use of WhatsApp Business-Enabled Conversational Commerce in Indonesia: A UTAUT2 Study

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The current paper considers the adoption and sustained utilization of WhatsApp Business-enabled conversational commerce by consumers in Indonesia based on the UTAUT2 model. While WhatsApp Business enjoys widespread use by organizations in their customer conversations, there is insufficient evidence as to whether consumer adoption of business messaging tools is made through deliberate intention or routine messaging activities in advanced mobile-first markets. A quantitative survey was conducted with 94 Indonesian participants, whose responses were analyzed with Partial Least Squares Structural Equation Modeling (PLS-SEM). The analysis revealed that habit is the primary determinant of both behavioral intention and actual use behavior, whereas performance expectancy has a notable impact on behavioral intention. Effort expectancy, social influence, hedonic motivation, price value, facilitating conditions, and behavioral intention–use behavior association, however, are insignificant. Overall, conversational commerce use in mature messaging environments seems to be motivated less by deliberate technology adoption intentions than routine communication activities. The current research adds to the existing body of knowledge by applying the UTAUT2 framework to the messaging-based conversational commerce environment and discovering an intention–behavior dissociation in business messaging systems.

Keywords—Conversational commerce; WhatsApp Business; UTAUT2; business messaging; habit.

I. BACKGROUND

Through digital transformation, organizations and customers now interact using platforms that allow real-time interactions through digital technology. In Indonesia, the preferred means of such communications is through messaging applications, with WhatsApp becoming popular among these. The growing use of messaging apps as tools of communication has resulted in the popularity of WhatsApp Business, which is currently considered a key conversational commerce platform. Using capabilities such as automated responses, chatbots, and real-time communication capabilities, WhatsApp Business provides businesses with a platform to communicate effectively with their customers and engage them in conversations [1], [5], [15].

Even though this application is widely used, current studies on WhatsApp Business are mainly concentrating on the business and marketing aspect, such as customer interaction and organizational effects [12], [14], [15]. However, there is a lack of investigation on the consumer's acceptance and use of this technology in their day-to-day

communication activities. Moreover, in the case of information systems research, prior literature does not analyze the system architecture behind such interactions, as they mostly depend on behavior-based models such as TAM or UTAUT.

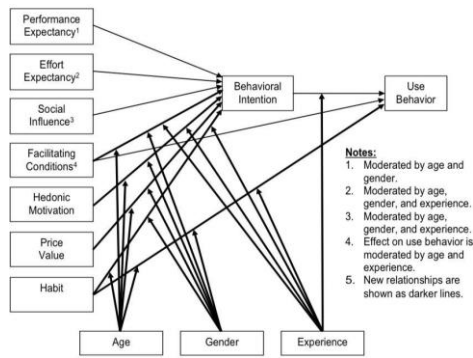
In order to fill the gaps, the current research implements the Unified Theory of Acceptance and Use of Technology 2 framework that comprises factors related to consumers, including hedonic motivation, price value, and habit [23]. Moreover, the chosen framework will be complemented by system architecture view on technology acceptance and usage. Combining both perspectives, this research attempts to provide a better insight into the issue under discussion.

II. LITERATURE REVIEW

A. Unified Theory of Acceptance and Use of Technology (UTAUT2)

The Extended Unified Theory of Acceptance and Use of Technology (UTAUT2) framework was selected as a theoretical perspective that "expands the original organizational theory to include consumer-related measures such as Hedonic Motivation, Price Value, and Habit" [23]. Constructs include:

- Performance Expectancy (PE): The amount to which WhatsApp Business assists in increasing the efficiency of customer interaction, or usefulness in terms of engaging with the customers.
- Effort Expectancy (EE): The level of easiness with which WhatsApp Business is used by the customer.
- Social Influence (SI): The influence of individuals surrounding the user, for example, friends or family members, that promote its usage.
- Facilitating Conditions (FC): Availability of resources that facilitate the use of WhatsApp Business.
- Habit (HB): User's behavioral habit to use WhatsApp Business depending on prior experiences using it [22].
- Hedonic Motivation (HM): The extent to which WhatsApp Business usage gives user pleasure or satisfaction.
- Price Value (PV): This factor refers to users' ability to weigh benefits of using WhatsApp Business in



relation to the price.

- Behavioral Intention (BI): It measures the probability that users will use WhatsApp Business consistently.
- Use Behavior (UB): Users' behavioral pattern in using WhatsApp Business's offered features.

Fig. 1. Research Model

B. WhatsApp Business as a Digital Communication Platform

The following features enable direct interaction between SMEs and their customers through WhatsApp Business:

- Business Profile: The creation of a profile with details of businesses, addresses, and business hours.
- Welcome Message/Automatic Reply Messages: Ensure that customers are informed about receiving their messages and provide them with automatic replies.
- Message Templates: Predefined messages about products or services.
- Management of Messages: Managing outgoing and incoming messages.
- Facebook Integration: Integrating messages on Facebook.

These features make it easy to establish personalized and time-bound communication, which could help increase customer satisfaction levels [10].

C. Previous Studies & Research Gap

Despite its increasing popularity in Indonesia, there are some limitations found in previous research on the topic of WhatsApp Business. The first limitation is from the empirical point of view, where research tends to concentrate on the business side rather than the consumer side of WhatsApp Business [12]. The study entitled "The Role of WhatsApp Business in Increasing Consumer Engagement" also highlights the impact of using WhatsApp Business by the businesses themselves to increase consumer growth [15]. On the other hand, research on how consumers perceive and utilize WhatsApp Business is rather scarce.

From a theoretical perspective, the literature on the acceptability of technology in the same field has mainly focused on theories like TAM (Technology Acceptance Model) and UTAUT (Unified Theory of Acceptance and Use of Technology). Even though these two theories are crucial, they tend to fail when it comes to incorporating consumer-related issues like enjoyment, perceived value, and habit. It is possible to address these three issues using the second version of UTAUT that consists of Hedonic Motivation, Price Value, and Habit [23]. Apart from this, there have been very few papers published on the architecture of the WhatsApp Business app. Most of them just discuss how people perceive the app.

The gaps identified in previous research have led this study to develop an expanded UTAUT model suited for the environment of WhatsApp Business, which involves the incorporation of Hedonic Motivation, Price Value, and Habit to enhance understanding of customer adoption and usage behaviors [16], [18]. Furthermore, a systems architecture view is introduced to account for the role technology plays in facilitating interaction.

D. Research Objective

This study aims to identify the factors driving consumer acceptance and continued use of WhatsApp Business as a conversational commerce platform by integrating UTAUT2 with a system architecture perspective. Specifically, the study investigates how Performance Expectancy, Effort Expectancy, Social Influence, Hedonic Motivation, Price Value, and Habit shape users' behavioral intention, while also examining the direct influence of Facilitating Conditions and Habit on actual use behavior. Additionally, the study explores whether behavioral intention translates into observable use, and identifies the most dominant adoption factors within the WhatsApp Business ecosystem.

III. RESEARCH METHODOLOGY

A. Research Design

In this research, the quantitative research method is employed through conducting a cross-sectional survey that provides an overall picture of the consumer adoption behaviors in relation to the selected research constructs at a certain point in time. The quantitative research method is appropriate since it enables researchers to measure several latent variables at once [21].

B. Population and Sampling

The population of interest for this research will be the consumers residing in Indonesia who have been engaged with a business organization through the use of WhatsApp Business application in the last three months. In order to conduct the survey on this population, a purposive sampling technique was adopted in order to identify relevant participants. It is estimated that 100 participants are sufficient for PLS-SEM analysis [21].

C. Research Instrument

Data will be gathered through an online survey with the help of Google Forms. Each question is based on the Likert Scale, with possible answers being either "Strongly Disagree" for 1 or "Strongly Agree" for 7. Each question is based on a pre-validated scale that has been employed in the UTAUT2 theory [23]. The construct and question counts include:

- Performance Expectancy (PE): 4 items (Benefits, efficiency).
- Effort Expectancy (EE): 4 items (Ease, clarity).
- Social Influence (SI): 3 items (Environmental influence, recommendations).
- Facilitating Conditions (FC): 4 items (Resources, knowledge).
- Hedonic Motivation (HM): 3 items (Pleasure, entertainment).
- Price Value (PV): 3 items (Value compared to data costs).
- Habit (HB): 4 items (Automation, routine).
- Behavioural Intention (BI): 3 items (Intention to continue using).

- Usage Behaviour (UB): 3-4 items (Frequency of use).

D. Data Analysis Technique

The Partial Least Squares Structural Equation Modeling (PLS-SEM) will be used in this study with the use of SmartPLS program. The method was selected due to the following reasons: it is effective for exploratory studies that incorporate multiple latent variables, provides accurate results when used on smaller sample size, and requires no specific distributional assumptions.

Analysis is done using two stages that are sequential. In the first stage, the measurement model (outer model) will be analyzed to validate the reliability and validity of constructs. Reliability will be measured using Cronbach's Alpha and Composite Reliability, values of which must be greater than 0.7. On the other hand, convergent validity will be assessed by calculating the Average Variance Extracted, whose value must not fall below 0.5.

The second phase will test the structural model (inner model) using the coefficient of determination (R^2) to determine the extent to which the variables can explain each other, and by computing the path coefficients through bootstrapping with 5,000 sub samples. The significance level for hypothesis testing will be one-tailed with a probability level of 0.05.

E. Research Hypothesis

This study proposes nine hypotheses grounded in the UTAUT2 framework. With respect to Behavioral Intention, Performance Expectancy is expected to positively influence users' intention to use WhatsApp Business, as consumers who perceive the platform as useful in improving their interaction with businesses are more likely to commit to its continued use (H1). Similarly, consumers who find the platform easy to operate are anticipated to show stronger intention to engage (H2), while those who perceive social encouragement from their environment are also expected to demonstrate higher intention (H3). Hedonic Motivation is hypothesised to strengthen intention through the enjoyment consumers derive from conversational interactions (H4), and Price Value is expected to contribute positively when users judge the platform's benefits to outweigh its costs (H5). Among all antecedents of Behavioral Intention, Habit is anticipated to exert the strongest influence, as repeated use becomes ingrained in daily communication routines (H6).

Regarding Use Behavior, Facilitating Conditions are hypothesised to support actual platform engagement by ensuring users have the necessary infrastructure and resources (H7), while Habit is expected to directly drive Use Behavior independent of conscious intention (H8). Lastly, Behavioral Intention is hypothesised to translate into observable use, consistent with established technology adoption theory (H9).

TABLE I. QUESTIONNAIRE TABLE

Questionnaire Table		
Code	Question	Ref
Performance Expectancy (PE)		
PE1	WhatsApp Business helps me communicate with businesses more quickly.	[23]
PE2	WhatsApp Business makes it easier for me to obtain product or service information.	[23]

PE3	WhatsApp Business increases my efficiency when interacting with businesses.	[23], [16]
PE4	WhatsApp Business improves the quality of my experience as a customer.	[23], [21]
Effort Expectancy (EE)		
EE1	WhatsApp Business is easy to learn.	[23]
EE2	Using WhatsApp Business does not require much effort.	[23]
EE3	The features of WhatsApp Business are easy to understand.	[23], [21]
EE4	WhatsApp Business is easy to use for communicating with businesses.	[23]
Social Influence (SI)		
SI1	People around me recommend using WhatsApp Business for business communication.	[23]
SI2	Many businesses I know use WhatsApp Business, which encourages me to use it.	[23], [16]
SI3	I use WhatsApp Business because it is a general trend in society.	[23]
Facilitating Conditions (FC)		
FC1	I have devices that adequately support the use of WhatsApp Business.	[23]
FC2	I have sufficient internet access to use WhatsApp Business.	[23]
FC3	The facilities I have make it easier for me to use WhatsApp Business.	[23], [21]
Hedonic Motivation (HM)		
HM1	Using WhatsApp Business to shop or inquire with businesses is enjoyable.	[23]
HM2	I enjoy interacting with sellers through WhatsApp Business.	[23], [16]
HM3	Using WhatsApp Business provides an engaging experience (e.g., using stickers/emojis, personal interactions, viewing a visual catalog).	[23]
Price Value (PV)		
PV1	The benefits of WhatsApp Business are worth the time I spend using it.	[23]
PV2	WhatsApp Business provides good value for me as a customer	[23], [21]
PV3	Using WhatsApp Business is more beneficial compared to other channels.	[23]
Habit (HB)		
HB1	I am accustomed to using WhatsApp Business in my daily activities.	[23], [22]
HB2	I frequently use WhatsApp Business for various communication needs.	[23]
HB3	Using WhatsApp Business feels automatic to me.	[23], [22]
Behavioral Intention (BI)		
BI1	I intend to continue using WhatsApp Business in the future.	[23]
BI2	I will always try to use WhatsApp Business in my daily business interactions.	[23]
BI3	I plan to use WhatsApp Business frequently in the near future.	[23], [21]
Use Behavior (UB)		
UB1	I frequently use WhatsApp Business to ask questions or request information.	[23], [21]
UB2	I use WhatsApp Business for transactions or consultations with businesses.	[23]
UB3	I prefer WhatsApp Business over other platforms for business communication.	[23]

IV. RESULT AND DISCUSSION

A. Descriptive Statistic Results

The results of the survey revealed that all the participants, 94 out of 94, use WhatsApp, with 96.8% of them having interacted with WhatsApp Business. The survey also revealed

that 54.25% of the participants were female, with the rest being male. The participants were of different ages, with the majority, 45.74%, being between 17-25 years, followed by those between 26-35 years, accounting for 27.65%. The majority of the participants had education above the undergraduate level, with different occupations.

B. Measurement Model (Outer Model)

The reliability and validity of the constructs were measured by using Cronbach's Alpha, Composite Reliability, and Average Variance Extracted. From Table I, it is clear that all constructs were found to be reliable and valid, with $\alpha > 0.7$, except for UB, which was 0.651, as well as $CR > 0.7$ and $AVE > 0.5$.

TABLE I. Construct Reliability and Validity

Construct	Cronbach's Alpha	Composite Reliability	AVE	Status
Behavioral Intention (BI)	0.913	0.946	0.853	Valid
Effort Expectancy (EE)	0.900	0.930	0.770	Valid
Facilitating Conditions (FC)	0.902	0.930	0.770	Valid
Habit (HB)	0.921	0.945	0.810	Valid
Hedonic Motivation (HM)	0.876	0.925	0.804	Valid
Performance Expectancy (PE)	0.900	0.930	0.769	Valid
Price Value (PV)	0.900	0.937	0.833	Valid
Social Influence (SI)	0.804	0.879	0.708	Valid
Use Behavior (UB)	0.651	0.801	0.604	Valid

C. Structural Model (Inner Model)

The structural model has been assessed using Coefficient of Determination (R^2), where R^2 for Use Behavior is 0.932 (Substantial), and R^2 for Behavioral Intention is 0.888 (Substantial). Hypothesis testing has been carried out by employing bootstrapping with 5,000 subsamples. The results are presented in Table II.

TABLE II. Hypothesis Testing Results

Hypothesis	Path	Path Coefficient (β)	P Value	Result
H1	PE $\rightarrow$ BI	0.304	0.029	Supported
H2	EE $\rightarrow$ BI	-0.395	0.102	Rejected
H3	SI $\rightarrow$ BI	0.069	0.259	Rejected

H4	HM $\rightarrow$ BI	0.211	0.109	Rejected
H5	PV $\rightarrow$ BI	0.041	0.808	Rejected
H6	HB $\rightarrow$ BI	0.746	0.000	Supported
H7	FC $\rightarrow$ UB	-0.075	0.165	Rejected
H8	HB $\rightarrow$ UB	0.750	0.000	Supported
H9	BI $\rightarrow$ UB	0.288	0.164	Rejected

D. Discussion

The findings point to a few important takeaways about how and why people use WhatsApp Business as a digital communication tool. Most notably, they highlight the central role of Habit in shaping how users adopt and use the platform.

The Dominance of Habit. Perhaps the most striking finding is just how powerfully Habit shapes user behavior. With a significant effect on both Behavioral Intention ($\beta = 0.746$) and Use Behavior ($\beta = 0.750$), the data suggests that people engage with WhatsApp Business not because they stop and think about it, but simply because it has become second nature. This echoes what prior research has found: once a technology settles into daily life, conscious decision-making gives way to automatic behavior [19].

From the technological standpoint, the implication of the finding is that the messaging-based business platforms are able to capitalize on the prevailing user behaviors associated with the use of messaging applications. Instead of users having to learn and get accustomed to the use of another system or application, the WhatsApp Business platform takes advantage of the messaging environment that users are already accustomed to using. The adoption of the platform is therefore made seamless for the users.

The Intention–Behavior Gap. Equally interesting is the non-significant relationship between Behavioral Intention and Use Behavior ($\beta = 0.288$, $p = 0.164$). This stands in contrast to what the UTAUT2 framework typically predicts, where intention is expected to be a strong predictor of actual use. Here, however, the two appear to operate independently, suggesting that people may use WhatsApp Business regardless of whether they consciously intend to. This points to a broader truth about mature platforms: when habit is strong enough, intention becomes almost irrelevant to what people actually do.

This has been seen in previous studies on information system continuance, where habit formation reduces the predictive power of behavioral intention [21]. With respect to WhatsApp Business, it is possible that people use the platform not because of their intention to use it, but because messaging using WhatsApp has simply become an integral part of their behavior.

In terms of system design, the findings highlight the importance of frictionless integration, where users can interact with business platforms through communication channels they already use and understand.

The Role of Performance Expectancy. The results also

show that Performance Expectancy has a significant positive relationship with Behavioral Intention ($p = 0.029$). This means users still consider how useful and effective WhatsApp Business is before deciding to use it as a communication platform.

This implies that people still consider the performance expectancy of WhatsApp Business before deciding to use it as a communication channel.

Previous studies on technology adoption consistently identify perceived usefulness as one of the strongest factors influencing user acceptance [22]. In the context of WhatsApp Business, users generally see the platform as a practical way to access information, communicate with businesses, and make interactions quicker and more efficient.

On the other hand, Effort Expectancy, Social Influence, and Price Value were found to have little impact on Behavioral Intention. One possible reason is that WhatsApp is already highly familiar to people in Indonesia. Because users are accustomed to the platform, ease of use is no longer a major consideration in adoption. Likewise, Social Influence becomes less relevant since WhatsApp usage is already widely accepted as part of daily communication. In mature markets, outside opinions and social pressure tend to have less influence on technology adoption [25], [26].

Building upon these findings, it becomes evident that the adoption of WhatsApp Business is not solely determined by user perceptions, but is also closely related to how the underlying technological system supports user interaction. The significance of constructs such as Habit and Performance Expectancy, as well as the diminishing role of other factors, suggests that seamless integration, familiarity, and system efficiency play a critical role in shaping user behavior. Based on this perspective, a conceptual architecture is proposed to explain how technological components support and strengthen these behavioral patterns.



Fig. 2. Proposed Conversational Commerce Architecture

The proposed architecture consists of several layers that collectively support conversational commerce interactions:

- User Layer represents the real customers who reach out to businesses through WhatsApp Business every day.
- Communication Layer enables real-time, two-way messaging through the WhatsApp platform, allowing conversations to happen seamlessly.
- Application Layer handles what happens behind the scenes by processing requests through chatbot systems, automated replies, and business logic to keep interactions running smoothly.
- Integration Layer bridges the application with the broader business ecosystem, connecting to external systems like CRM platforms and backend services

through APIs.

- Data Layer stores all interaction data and drives the analytics that help businesses learn from each conversation and make smarter decisions over time.

This architecture demonstrates that UTAUT2 constructs are supported by underlying technological systems rather than operating independently. Performance Expectancy is shaped by how efficiently the application and integration layers operate, while Effort Expectancy reflects how naturally the communication and application layers simplify the experience for users. Social Influence emerges from widespread adoption within the user ecosystem, and Facilitating Conditions are enabled by the supporting infrastructure in the integration and data layers. Meanwhile, Hedonic Motivation, Price Value, and Habit are shaped by the overall user experience across the system. This shows that user acceptance and usage of WhatsApp Business are not only influenced by perceptions but also by the effectiveness of the underlying system architecture.

Overall, these findings suggest that in a mature market, traditional adoption factors such as ease of use and social influence may no longer play a major role. Instead, behavioral factors, particularly habit, become more dominant in influencing technology adoption. From a technology development perspective, this means that integrating with existing communication habits is more important than introducing entirely new interaction methods. The findings show that habit is the most influential factor in adoption, indicating that the design of conversational commerce platforms like WhatsApp Business has a stronger impact on user behavior than traditional e-commerce platform design.

V. CONCLUSION

The present study investigated the determinants of consumer acceptance and adoption of WhatsApp Business application as a conversational commerce solution utilizing UTAUT2 framework. It was found that Habit is the highest influencing determinant for both usage intention and usage of the application, emphasizing how regular messaging practices have an impact on consumers' interactions with businesses. Performance Expectancy also had a significant influence on Behavioral Intention, while several traditional technology adoption factors showed limited impact in this context. These results suggest that conversational commerce adoption in messaging platforms is strongly influenced by habitual communication practices, highlighting how important it is for business messaging systems to fit naturally into the ways people already communicate.

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